JULFIKER IBN HAIDER

Mechatronics Engineer | Embedded Systems and IoT

Portfolio

PROFILE

A Mechatronics & Industrial Engineering graduate from CUET with hands-on experience in embedded systems, IIoT product development, and industrial automation. Currently working at **Southern IoT** as an Implementation Lead & Hardware Designer, spearheading end-to-end IIoT deployments for industrial and healthcare sectors. Skilled in firmware development, LPWAN communication, RS485/Modbus protocols, CNC machine fabrication, and IoT cloud infrastructure. Strong leadership background through CUET's core technical robotics club.

EDUCATION

B.Sc. Mechatronics & Industrial Eng.
CUET
2020 – 2025 | CGPA: 3.10

HSC – Science
Tamirul Millat Kamil Madrasah
2017 – 2019 | GPA: 5.00

SSC – Science
Chaprashirhat Fazil Madrasah
2012 – 2017 | GPA: 4.78

WORK EXPERIENCE

EMBEDDED SYSTEMS & IOT ENGINEER
Southern IoT
2025 – Present

Implementation Lead · Hardware Designer · Firmware Developer

- Leading hardware design and firmware development for IIoT products deployed across industrial sites.
- Engineered data acquisition modules using RAK3173 (LoRaWAN) and ESP32/ESP8266 with RS485/RS232 (Modbus RTU/ASCII) communication.
- Designed and fabricated a CNC Router for precision workpiece cutting and a Laser CNC Machine for engraving operations.
- Developed an Automatic Wire Bending Machine, automating wire-form production and improving repeatability.
- Established a "Standard Installation Method" toward a One-Tap Installation philosophy, cutting deployment time significantly.
- Led industrial enclosure design balancing ruggedness, cost-efficiency, and brand aesthetics.

GENERAL SECRETARY
Robo Mechatronics Association (RMA), CUET
2022 – 2024

- Served as Executive Member and team lead for Expeditious-2023.
- Organized workshops on IoT, Robotics, and Microcontrollers for junior batches at CUET and partner institutions.

CONTACT

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CORE SKILLS

- Embedded Systems
- IoT / IIoT Development
- Firmware (C/C++/Python)
- RS485 / Modbus Protocols
- PCB & Enclosure Design
- CNC Machine Building
- ROS 2 / Robotics
- PLC / Industrial Automation

TECHNOLOGIES

ESP32/ESP8266
RAK3112/RAK3172 (LORAWAN)
Raspberry Pi 4/5 · MQTT
Node-RED · InfluxDB · Grafana
Modbus RTU/ASCII
ROS 2 · SolidWorks · TIA Portal
MATLAB · Proteus · Linux
FreeRtos · GRBL / G-code

LANGUEAGE

Bangla ★★★★★
English ★★★★
Arabic ★★

RELEVANT COURSEWORK

Microcontroller & Interfacing
Control Systems
Power Electronics
Machine Learning
Sensor & Instrumentation
Digital Logic Design
Industrial Automation
FreeRtos · GRBL / G-code
CIM · CAD/CAM · Robotics

PROJECTS

★ CORE PRODUCT · SOUTHERN IOT

CAREFLOW V1 - INDUSTRIAL IOT & ENERGY MONITORING SYSTEM
Implementation Lead & Hardware Designer
2025 – Present

An end-to-end IIoT solution for real-time monitoring of industrial generators and Rural Electrification Board (REB) power lines. Bridges legacy industrial machinery with cloud-based analytics using LPWAN and standard industrial protocols.

- Industrial IoT & Energy Monitoring System
- Implementation Lead & Hardware Designer
- An end-to-end IIoT solution for real-time monitoring of industrial generators and Rural Electrification Board (REB) power lines. Bridges legacy industrial machinery with cloud-based analytics using LPWAN and standard industrial protocols.
- Engineered firmware for RAK3173 & ESP32/ESP8266 to fetch telemetry via RS485/RS232 (Modbus RTU/ASCII).
- Led site inspections and machine compatibility testing as primary technical lead.
- Standardized field deployment process; designed rugged industrial enclosures for harsh environments.
- Managed edge-to-server connectivity ensuring high uptime for critical energy data transmission.

HOSPITAL MANAGEMENT & WORKFLOW SYSTEM
Project Manager · Southern IoT

A comprehensive healthcare digitization platform optimizing clinical workflows and patient data management, scaled from a single-department pilot to a hospital-wide implementation.

- Automated patient admission, discharge, and inter-department transfers with consistent data integrity.
- Built centralized EHR database architecture with HIPAA-aligned security protocols.
- Developed APIs connecting Pharmacy, Pathology, and Radiology departments into a single source of truth.
- Designed a high-density clinical dashboard to reduce cognitive load for medical practitioners.

CNC ROUTER & LASER CNC MACHINE
Designer & Builder · Southern IoT

Designed and built two CNC machines from scratch — a CNC Router for precise workpiece dimensioning and a Laser CNC for engraving and cutting operations.

AUTOMATIC WIRE BENDING MACHINE
Designer & Developer · Southern IoT

Developed an automated wire bending machine capable of forming wires into predefined shapes with high repeatability, minimizing manual intervention in production.

UNDERGRAD PROJECTS

EV MOTOR FAULT DETECTION & HEALTH MONITORING SYSTEM
Lead Researcher · Undergraduate Thesis, CUET

Portable multi-sensor system using Raspberry Pi 4 & ESP32 for real-time EV motor monitoring with cloud analytics pipeline and custom PCB design.

AUTONOMOUS MOBILE ROBOT — 3D SLAM & NAVIGATION
Pre-Defence Project · CUET

ROS 2-based Simulation for autonomous robot for 3D SLAM and navigation, designed as a functional office desk delivery robot.

INDUSTRIAL OBJECT PICK & PLACE ROBOT
4-DOF robotic arm with Pi Camera for object detection and automated placement.

IOT-BASED IPS MONITORING SYSTEM
Real-time power monitoring and control of IPS units with IoT relay management.

TRAINING & INDUSTRIAL EXPOSURE

INDUSTRIAL AUTOMATION COURSE
IICT, CUET & BCC, ICT Division — 60 Hours

- FluidSIM, TIA Portal, PLC Ladder Logic Programming, HMI Design, Pneumatic & Electro-Pneumatic Systems.

INDUSTRIAL ATTACHMENT — WALTON HI-TECH INDUSTRIES PLC — **JAN 2025 · 14 DAYS**
GAZIPUR

- Injection/extrusion molding, PU foaming, thermoforming, PCB & cable production, AC/refrigerator assembly & QC testing.

BASIC SAFETY TRAINING
SAMSUNG C&T CORPORATION

- First Aid, Fire Safety, PST, PSSR, Emergency Procedures.

SOLIDWORKS CERTIFICATION

SOLIDWORKS 3D CREATOR ASSOCIATE

SOLIDWORKS CAD Design Associate

REFERENCES

Sanjeeb Roy
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